

EXECUTIVE SUMMARY: ABC Inventory Classification Analysis

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Subject: Strategic Inventory Categorization based on the Pareto Principle (80/20 Rule)

1. Objective

The purpose of this analysis is to classify the current warehouse inventory into three strategic categories (A, B, and C) based on their annual financial impact. This classification enables the Supply Chain and Operations teams to prioritize resources, optimize physical storage, and implement targeted procurement strategies.

2. Methodology

The inventory was evaluated using the **ABC Classification Model**, which applies the Pareto Principle. The total annual value for each SKU was calculated by multiplying the unit cost by the annual units sold. The products were then sorted in descending order of their total financial value and categorized based on their cumulative percentage of the total warehouse value (\$159,750):

- **Class A:** Items representing approximately the top 80% of the total inventory value.
- **Class B:** Items representing the next 15% (up to 95% cumulative value).
- **Class C:** The remaining items comprising the bottom 5% of the total value.

3. Key Findings

The analysis successfully segmented the 8 evaluated SKU groups into the following categories:

- Class A (High Financial Impact - 77.62% of Total Value):
 - This category includes **Laptops, Monitors, and Headphones**.
 - Despite representing only a portion of the physical units, these three items generate \$124,000 in annual value.
 - *Operational Status:* Critical. Any stockout in these SKUs represents a severe financial risk.
- Class B (Intermediate Impact - 15.02% of Total Value):
 - This category includes **keyboards, Mouse, and Ergonomic Chairs**.

- These items bring the cumulative financial value up to 92.64%.
- *Operational Status:* Stable. These items require regular monitoring but less aggressive control than Class A.
- Class C (Low Financial Impact - 7.36% of Total Value):
 - This category includes **Webcams and HDMI cables**.
 - Notably, Cables HDMI have the highest sales volume (1,000 units annually) but contribute the least to the total financial value (\$5,000).
 - *Operational Status:* Low priority for capital allocation, but potentially high impact on physical warehouse space.

4. Strategic Recommendations

Based on the classification, the following operational adjustments are recommended to management:

1. **Strict Control for Class A:** Implement continuous inventory reviews (cycle counting) and tighter safety stock policies for Laptops and Monitors to prevent stockouts and protect capital.
2. **Automated Replenishment for Class C:** Transition HDMI cables and Webcams to automated, bulk-purchasing models with less frequent physical audits to save labor hours.
3. **Space Optimization:** Relocate Class A items to the most accessible warehouse zones (forward picking areas) to reduce travel time for warehouse operators, as these are high-value priority items.